



BOARD QUESTION PAPER : MARCH 2014

Note:

- All questions are compulsory.
- Figures to the right indicate full marks.
- Graph papers are not necessary for L.P.P. Only rough sketch of the graph is expected.
- Answers to both the sections should be written in the separate answer books.
- Answer to every new question must be written on a new page.

Section – I

Q.1. Attempt any SIX of the following:**[12]**

- If $A = \begin{bmatrix} 1 & 2 & -3 \\ 5 & 4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 & 3 \\ -2 & 5 & 0 \end{bmatrix}$
then find $2A + 3B$. (2)
- If the function f is continuous at $x = 1$, then find $f(1)$.
Where $f(x) = \frac{x^2 - 3x + 2}{x - 1}$ for $x \neq 1$. (2)
- If $x = \tan^{-1}t$ and $y = t^3$, find $\frac{dy}{dx}$. (2)
- Evaluate: $\int \sin^2 x \, dx$. (2)
- Write negation of the following statements:
 - Chetan has black hair and blue eyes.
 - $\exists x \in \mathbb{R}$ such that $x^2 + 3 > 0$. (2)
- If $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then find $|AB|$. (2)
- Evaluate: $\int \frac{dx}{4 - 9x^2}$ (2)
- If the function f is continuous at $x = 2$, then find 'k'
where $f(x) = \frac{x^2 + 5}{x - 1}$, for $1 < x \leq 2$
 $= kx + 1$, for $x > 2$ (2)

Q.2. (A) Attempt any TWO of the following:**[6][14]**

- If $x^y = e^{x-y}$, show that $\frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}$ (3)
- If $\sin y = x \sin(a + y)$
prove that $\frac{dy}{dx} = \frac{\sin^2(a + y)}{\sin a}$ (3)
- Discuss extreme values of the function $f(x) = x \log x$. (3)

**(B) Attempt any TWO of the following:**

- i. Discuss the continuity of the function
- f
- at
- $x = 0$
- ,

$$\text{where } f(x) = \frac{5^x + 5^{-x} - 2}{\cos 2x - \cos 6x}, \quad \text{for } x \neq 0$$

$$= \frac{1}{8}(\log 5)^2, \quad \text{for } x = 0 \quad (4)$$

- ii. The expenditure
- E_C
- of a person with income
- I
- is given by
- $E_C = (0.000035) I^2 + (0.045) I$
-
- Find marginal propensity to consume (MPC) and average propensity to consume (APC)
-
- when
- $I = 5000$
- . (4)

- iii. Evaluate:
- $\int x \cot^{-1} x \, dx$
- (4)

Q.3. (A) Attempt any TWO of the following:**[6][14]**

- i. If
- p
- : It is a day time.

 q : It is warm.

Given the verbal statements for the following symbolic statements:

a. $p \wedge \sim q$

b. $p \vee q$

c. $p \leftrightarrow q$

(3)

- ii. Using the truth table, examine whether the statement pattern
- $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$
- is a tautology, a contradiction or a contingency. (3)

- iii. The cost
- C
- of producing
- x
- articles is given as

$C = x^3 - 16x^2 + 47x.$

For what values of x will the average cost be decreasing? (3)**(B) Attempt any TWO of the following:**

- i. If
- $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix}$
- then find
- A^{-1}
- by using elementary transformation. (4)

- ii. Evaluate:
- $\int_0^3 \frac{dx}{x + \sqrt{9 - x^2}}$
- (4)

- iii. Find the volume of a solid obtained by the complete revolution of the ellipse
- $\frac{x^2}{36} + \frac{y^2}{25} = 1$
-
- about
- X
- axis. (4)

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- All questions are compulsory.
- Figures to the right indicate full marks.
- Answer to every question must be written on a new page.
- L.P.P. problem should be solved on graph paper.
- Log table will be provided on request.
- Write answers of Section – I and Section – II in one answer book.

Section – I

Question 1 to 3 (based on section I) are given in our book **STD XII (COMMERCE) MATHEMATICS AND STATISTICS - I**

Section – II**Q.4. Attempt any SIX of the following:****[12]**

- Alex spends 20% of his income on food items and 12% on conveyance, If for the month of June 2010, he spent ₹ 900 on conveyance, find his expenditure on food items during the same month. (2)
- Find the premium on a property worth ₹ 12,50,000 at 3% if the property is fully insured. (2)
- The following table gives the age of the husbands and of the wives:

Age of wives (in years)	Age of husbands (in years)			
	20-30	30-40	40-50	50-60
15 – 25	5	9	3	–
25 – 35	–	10	25	2
35 – 45	–	1	12	2
45 – 55	–	–	4	16
55 – 65	–	–	–	4

- Find the marginal frequency distribution of the age of husbands. (2)
- For a bivariate data $\bar{x} = 53$, $\bar{y} = 28$, $b_{YX} = -1.5$, $b_{XY} = -0.2$. Estimate Y, when X = 50. (2)
- Values of two regression coefficients between the variables X and Y are $b_{YX} = -0.4$ and $b_{XY} = -2.025$ respectively. Obtain the value of correlation coefficient. (2)
- Verify whether the following function can be regarded as probability mass function (p.m.f.) for the given values of X: (2)

X	-1	0	1
P (X = x)	-0.2	1	0.2

- The p.m.f. of a random variable X is

$$P(x) = \frac{1}{5}, \text{ for } x = 1, 2, 3, 4, 5$$

$$= 0, \text{ otherwise}$$

Find E (X).

(2)



- viii. The time (in hours) required to perform the printing and binding operations (in that order) for each book is given in the following table:

Books	I	II	III	IV	V
Printing Machine M_1	3	7	4	5	7
Binding Machine M_2	6	2	7	3	4

Find the sequence that minimizes the total elapsed time (in hours) to complete the work. (2)

Q.5. (A) Attempt any TWO of the following: (6)[14]

- i. Find the present value of an annuity immediate of ₹ 18,000 p.a. for 3 years at 9% p.a. compounded annually. [Given $(1.09)^{-3} = 0.7722$] (3)
- ii. Compute rank correlation coefficient for the following data:

R_x	1	2	3	4	5	6
R_y	6	3	2	1	4	5

(3)

- iii. If the rank correlation coefficient is $\frac{2}{3}$ and $\sum d_i^2 = 55$, then find the number of pairs of observations. Assume that no rank is repeated. (3)

(B) Attempt any TWO of the following:

- i. From the following data, find crude death rates (C.D.R.) for Town I and Town II, and comments on the results:

Age group (years)	Town I		Town II	
	Population	No. of deaths	Population	No. of deaths
0 – 10	1500	45	6000	150
10 – 25	5000	30	6000	40
25 – 45	3000	15	5000	20
45 and above	500	22	3000	54

(4)

- ii. Calculate the quantities indicated by ‘?’ for the following part of a life table:

x	l_x	d_x	q_x	L_x	T_x	e_x^0
4	9100	60	?	?	510000	?
5	?	45				

(4)

- iii. The Probability that a bomb dropped from an aeroplane will strike a target is $\frac{1}{5}$. If four bombs are dropped, find the probability that
- a. exactly two will strike the target.
- b. at least one will strike the target. (4)

Q.6. (A) Attempt any TWO of the following: (6)[14]

- i. Amit and Rohit started a business by investing ₹ 20,000 each. After 3 months Amit withdrew ₹ 5,000 and Rohit put in ₹ 5,000 additionally. How should a profit of ₹ 12,800 be divided between them at the end of the year? (3)
- ii. A bill of ₹ 7,500 was discounted for ₹ 7,290 at a bank on 28th October 2006. If the rate of interest was 14% p.a., what is the legal due date of the bill? (3)



- iii. Let X be the number of matches played by the player and Y be the number of matches in which he scored more than 50 runs. The following data shown is obtained for 5 players:

No. of Matches Played (X)	Data of matches of 5 players				
	21	25	26	24	19
Scored more than 50 in a match (Y)	19	20	24	21	16

Find the regression line of X on Y .

(3)

(B) Attempt any TWO of the following:

- i. Find the sequence that minimizes total elapsed time (in hours) required to complete the following jobs on two machines M_1 and M_2 in the order $M_1 - M_2$. Also find the minimum elapsed time T and idle times for the two machines.

(4)

Jobs Machines	A	B	C	D	E
M_1	5	1	9	3	10
M_2	2	6	7	8	4

- ii. Solve the following L. P. P. :

Minimize : $Z = 4x + 2y$

Subject to : $3x + y \geq 27$,

$x + y \geq 21$,

$x + 2y \geq 30$,

$x \geq 0, y \geq 0$

(4)

- iii. Solve the following L. P. P.:

Maximize : $Z = 4x + 10y$

Subject to : $2x + 5y \leq 10$

$5x + 3y \leq 15$

$x \geq 0, y \geq 0$

(4)